

MOHAMAD IKHWAN BIN JOHAR

Mobile Application Developer | Flutter | IoT Kiosk Apps | WebSocket | Payment and API Integration

Malaysia | 01141312421 | ikhwanj95@gmail.com | sicx6.github.io/Ikhwan-Portfolio/ | github.com/Sicx6 |
linkedin.com/in/mohamad-ikhwan-935889214

PROFESSIONAL SUMMARY

Mobile Application Developer with nearly 3 years of Flutter experience building smart fridge kiosk applications, customer mobile apps, Flutter Web/admin tools, and CodeIgniter 3 backend APIs. Experienced in GetX, REST API integration, WebSocket workflows, SQLite local storage, dual-mode HID keyboard and serial scanner integration, Firebase Cloud Messaging, Firebase Crashlytics, and payment integration. Also developing a Flutter-based AI-assisted audio analysis prototype connected to a local Python service.

TECHNICAL SKILLS

Mobile Development: Flutter, Dart, Android SDK, responsive UI, Android kiosk applications, Flutter Web, Figma-to-Flutter implementation
State Management and Architecture: GetX (primary), Riverpod, Provider, feature-based structure, MVVM, repository pattern, controller/service separation

Backend and Web: CodeIgniter 3, PHP, RESTful APIs, JavaScript, HTML5, CSS3, Bootstrap, AJAX

Database and Storage: MySQL, SQLite/sqlite, MongoDB, local caching, offline data persistence

Real-Time and Device Integration: WebSocket, MQTT exposure, HID keyboard scanner input, serial communication, hardware telemetry, edge-device integration

Firebase / Payments / Tools: Firebase Cloud Messaging, Firebase Crashlytics, RMS and QR payments, e-receipt API, Git, GitHub, Postman, Android Studio

Additional Exposure: Python, FFmpeg, yt-dlp, AI/model integration

PROFESSIONAL EXPERIENCE

Mobile Application Developer - Nuvening | 12/2023 - Present

- Built and maintained Flutter applications across Android kiosk, customer mobile, and Flutter Web/admin use cases for smart fridge and custom automated vending solutions.
- Used GetX as the primary state-management approach for the AI Smart Fridge application, organizing controllers and state for machine configuration, payments, transactions, refill operations, remote access, and operational modules.
- Implemented SQLite/sqlite local storage for machine configuration, payment options, products, credentials, sales, refunds, and operational data required by the smart fridge kiosk application.
- Implemented dual-mode barcode/QR scanner support for the AI Smart Fridge using HID keyboard input and serial communication, allowing the application to work with different scanner configurations.
- Translated approved Figma designs into responsive Flutter screens and reusable UI components while preserving intended layouts and interaction flows.
- Developed smart fridge kiosk flows involving scanner input, payment state handling, door/open-session logic, transaction summaries, refill workflows, and user-facing error handling.
- Integrated REST APIs, including e-receipt workflows that retrieve and present digital receipt links for completed transactions.
- Implemented real-time WebSocket communication between Flutter clients and smart fridge edge services for product-detection events, refill sessions, payment confirmation, transaction updates, and machine-state handling.
- Integrated Firebase Cloud Messaging and Crashlytics, plus QR/RMS payment flows covering deposits, payment confirmation, refunds, local persistence, and recovery scenarios.
- Built and maintained CodeIgniter 3/PHP REST APIs and contributed to MySQL database design for machine operations, remote access approval, payments, refill records, transaction records, product datasets, and administration.
- Developed admin portal modules using PHP, JavaScript, Bootstrap, Select2, SweetAlert, Dropzone, and AJAX for machine setup, payment credentials, product ownership, planograms, media uploads, and transaction review.

Protegee - Admin Assistant - Sime Darby Motors | 09/2022 - 01/2023

- Supported aftersales and sales operations through fleet/order documentation, invoices, filing, scanning, Microsoft Office/Outlook tasks, and internal record management.

IT Intern - Netherlands Maritime University College (NMUC) | 10/2021 - 02/2022

- Provided IT support for servers, CCTV, printers, PCs, network equipment, university systems, website support, troubleshooting, and technical documentation.

ADDITIONAL EARLIER EXPERIENCE

Assistant Manager - 7-Eleven Malaysia | 11/2017 - 01/2018 - Store operations, inventory, cashiering, administration, staff coordination, customer service, and reporting.

Graphic Designer - Dik Print | 03/2017 - 05/2017 - Designed banners, buntings, logos, T-shirts, stickers, and print materials.

Trainee Technician - Portal One Sdn Bhd | 10/2015 - 05/2016 - Computer maintenance, hardware troubleshooting, equipment setup, and client IT support.

SELECTED PROJECTS & ACHIEVEMENTS

PROFESSIONAL PROJECTS

AI Smart Fridge Ecosystem | Flutter, GetX, SQLite, CodeIgniter 3, MySQL, WebSocket, RMS, Firebase

- Used GetX as the primary state-management solution across machine state, payments, transactions, refill operations, remote access, configuration, and application workflows.
- Implemented SQLite local persistence and dual-mode barcode/QR scanner input using HID keyboard mode and serial communication.
- Developed mobile and kiosk modules for remote door access, AI refill, product image/dataset upload, transaction records, payment confirmation, machine configuration, and owner approval workflows.
- Connected Flutter UI, REST APIs, SQLite, payment services, WebSocket communication, and edge inference events for end-to-end smart fridge operations.

Custom Vending Machine Applications | Flutter, REST APIs, Figma

- Developed and enhanced Flutter application screens and transaction workflows for custom vending projects, focusing on application-side functionality and backend API integration.
- Translated Figma designs into responsive Flutter interfaces with reusable widgets and interaction states, and integrated receipt/e-receipt APIs into completed transaction workflows.

PERSONAL PROJECTS

AI Chord Finder | Flutter, Dart, Python, FFmpeg, yt-dlp

- Developed a Flutter chord-practice prototype that sends a YouTube URL to a local Python service and converts analyzed audio into a timed chord sequence.
- Integrated a Python analysis backend using yt-dlp for audio acquisition, FFmpeg for media conversion, and the BTC-ISMIR19 model for chord recognition.
- Built the Flutter-to-backend analysis flow and structured the repository to exclude generated audio, model dependencies, local environments, credentials, and other development artifacts from source control.

Advanced E-Commerce Application | Flutter, Node.js, MongoDB

- Developed a mobile shopping application with product browsing, cart, checkout, backend integration, and an admin inventory panel.

Re-Cycle Mobile App | Flutter, Local Storage, Geolocation

- Created a green-tech mobile application for recycling milestone tracking and nearby drop-off location discovery.

EDUCATION

Bachelor of Information Technology (Hons.) - Universiti Teknologi MARA (UiTM) | CGPA: 3.16 | 2018 - 2022
Diploma in Computer Science - Kolej Universiti Poly-Tech MARA (KUPTM) | CGPA: 2.84 | 2013 - 2016

LANGUAGES

Malay: Native / Fluent | English: Professional Working Proficiency

CERTIFICATIONS

The Complete 2021 Flutter Development Bootcamp with Dart (2022) | Dart Programming Certification - Programming Hub (2022) | Huawei HCIA Datacom (2022) | Hybrid Mobile Application Using Flutter Workshop (2021) | Usability Testing: SUS Tool Workshop (2021) | MUET - Band 4 (2016)